

AHB Master FSM - State Table

AHB Master Finite State Machine

State Encoding

State	Binary	Description
IDLE	000	Default - No transaction
ADDR_PHASE	001	Address phase - Send address to bus
DATA_PHASE	010	Data phase - Transfer data
WAIT_STATE	011	Wait for slave ready
BURST	100	Burst transfer in progress
ERROR_STATE	101	Error response received

Table 1: State Encoding

State Transition Table

Current State	Condition	Next State
IDLE	enable = 1 enable = 0	ADDR_PHASE IDLE
ADDR_PHASE	Always (after 1 cycle)	DATA_PHASE
DATA_PHASE	HRESP = 1 HREADY = 0 HREADY = 1 & burst_mode = 1 HREADY = 1 & burst_mode = 0	ERROR_STATE WAIT_STATE BURST IDLE
WAIT_STATE	HRESP = 1 HREADY = 1 & burst_mode = 1 HREADY = 1 & burst_mode = 0 HREADY = 0	ERROR_STATE BURST IDLE WAIT_STATE
BURST	HRESP = 1 HREADY = 0 HREADY = 1 & burst_count = 3 HREADY = 1 & burst_count < 3	ERROR_STATE WAIT_STATE IDLE BURST
ERROR_STATE	Always	IDLE

Table 2: State Transition Table

Output Signals Per State

State	HTRANS	HADDR	HWDATA	HWRITE
IDLE	00	0	0	0
ADDR_PHASE	10	in_HADDR	0	in_HWRITE
DATA_PHASE	10	addr_reg	wdata_reg (if write)	hwrite_reg
WAIT_STATE	10/11	addr_reg/burst_addr	wdata_reg (if write)	hwrite_reg
BURST	11	burst_addr	wdata_reg (if write)	hwrite_reg
ERROR_STATE	00	0	0	0

Table 3: Output Signals Per State

HBURST Value	Type	Beats
3'b010	WRAP4	4
3'b011	INCR4	4

Table 4: Burst Types Supported

Burst Support

Simplified State Diagram Summary

State	What Happens
IDLE	Wait for enable to start
ADDR_PHASE	Send address and control signals (1 cycle)
DATA_PHASE	First data transfer phase
WAIT_STATE	Slave is slow → Wait for HREADY=1
BURST	Multiple sequential transfers
ERROR_STATE	Slave returned error → Go back to IDLE

Table 5: Quick Reference - State Descriptions

Flow Hierarchy

Transaction Type	Path
Single Transfer	IDLE → ADDR_PHASE → DATA_PHASE → IDLE
Burst Transfer	IDLE → ADDR_PHASE → DATA_PHASE → BURST → BURST → ... → IDLE
With Wait State	... → DATA_PHASE → WAIT_STATE → BURST/IDLE
With Error	... → DATA_PHASE/WAIT_STATE/BURST → ERROR_STATE → IDLE

Table 6: Transaction Flow Hierarchy

Key Signals Description

Signal	Description
enable	Master wants to start a transaction
HREADY	Slave is ready (1) or not ready (0)
HRESP	Slave response: 0 = OK, 1 = ERROR
burst_mode	1 = Burst transfer, 0 = Single transfer
burst_count	Tracks burst beats (0 to 3 for 4-beat burst)

Table 7: Key Signals Description